

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS:

Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
B. `<video src="video.mp4" controls></video>`

- C. <movie src="video.mp4"></movie>
- D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. <div>
- B. <nav>
- C. <section>
- D. <header>

Snippet 4: XML Syntax

- A. <note><to>User</to><from>Admin</from>
- B. <note><to>User</to><from>Admin</from></note>
- C. <note><to>User</from></note>
- D. <note to="User"></note>

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

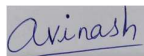
Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I Avinash k(192411044) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



1. Video Embedding — Answer: B. `<video src="video.mp4" controls></video>`

`<video>` is the standard HTML5 element specifically designed for embedding video content. It's natively supported by all modern browsers, supports the controls attribute for playback UI, and allows fallback content, multiple `<source>` tags, and attributes like autoplay, loop, and muted. Options A (`<media>`), C (`<movie>`), and D (`<vid>`) are not valid HTML elements — browsers won't recognize them at all.

2. Form Submission Method — Answer: B. POST

Registration data must be sent securely and can include sensitive information (passwords, personal details). POST sends data in the request body rather than the URL, so it isn't exposed in browser history, server logs, or bookmarks. It also has no length restrictions, unlike GET.

- GET (A): appends data to the URL as query parameters — insecure and unsuitable for sensitive data.
- PUT (C): used to update/replace an existing resource, not create a new registration.
- DELETE (D): used to remove a resource — irrelevant to form submission.

3. Navigation Element — Answer: B. `<nav>`

`<nav>` is a semantic HTML5 element specifically meant to wrap major navigation links (menus, tables of contents, pagination). It improves accessibility (screen readers can identify and jump to it) and SEO (search engines understand its purpose).

- `<div>` (A): purely structural, carries no semantic meaning.
- `<section>` (C): represents a generic thematic grouping of content, not navigation specifically.
- `<header>` (D): represents introductory content for a page or section — it may contain a `<nav>`, but isn't itself a navigation element.

4. Correct XML Syntax — Answer: B. `<note><to>User</to><from>Admin</from></note>`

Well-formed XML requires every opening tag to have a matching closing tag, and the entire document must have a single root element that is properly closed.

- A: missing the closing `</note>` root tag — incomplete.
- C: mismatched tags — opens `<to>` but closes with `</from>`, violating proper nesting.
- D: the closing tag `</note>` is malformed (missing the closing `>`), making the XML invalid.

Only B has correctly nested, matched, and closed tags.

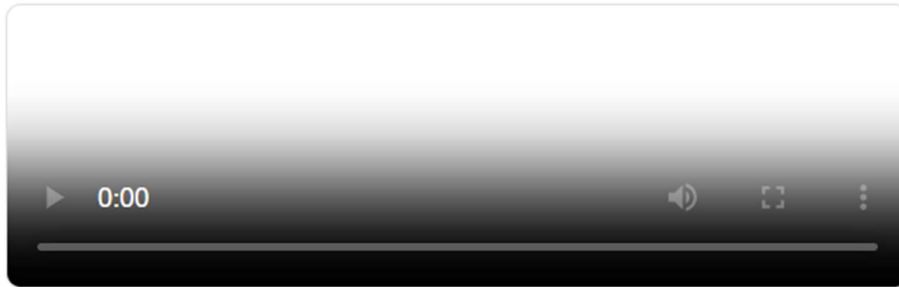
5. Web Communication Protocol — Answer: B. HTTP

HTTP (HyperText Transfer Protocol) — and its secure variant HTTPS — is the foundational protocol for transmitting web pages, form data, and API requests/responses between browsers and servers.

- FTP (A): used for transferring files, not for browser-server web page communication.
- SMTP (C): used for sending email, not web traffic.
- SNMP (D): used for network device management/monitoring, unrelated to web communication.

<https://avinash19107.github.io/CSA4301-IP/>

Promotional Video



Registration

Full name

Email

Password

Create account

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand